RAGHAV MALIK

Hardware Systems & R&D Technical Lead

Multidisciplinary R&D leadership · Electronics · RF · Mechatronics · Validation & field deployment

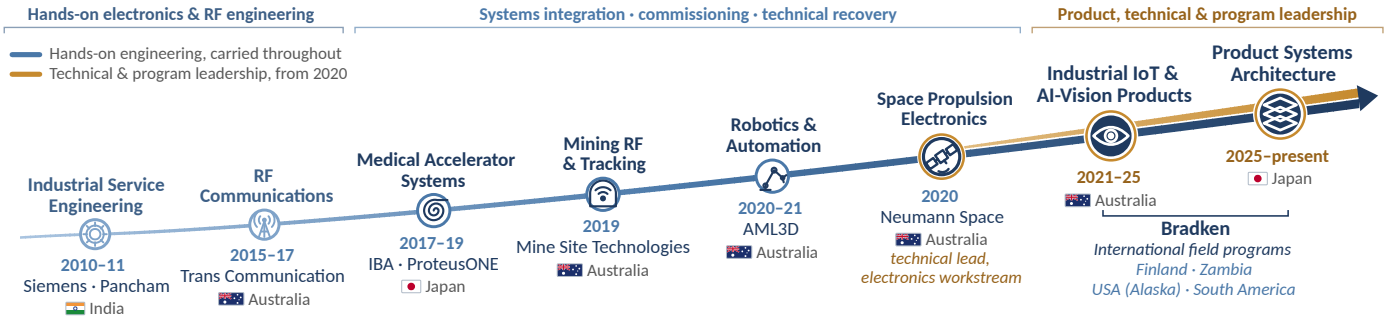
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PROFILE

Hardware systems engineer who grew into multidisciplinary R&D technical leadership. Over 12+ years, he has built, integrated and deployed complex physical systems: mining IoT and AI-vision products, space-propulsion electronics, a proton-therapy accelerator, a robotic metal-additive platform and underground RF communications.

His technical leadership starts in the lab. His hands-on depth in electronics, RF and mechatronics (bring-up, RF characterization, reproducing failures, finding root cause) is the basis for his architecture and schedule decisions, his reviews of supplier designs, the recovery of stalled programs and the commitments he makes to customers. He most recently served as technical and program lead, or principal hardware specialist, across five parallel mining R&D programs at **Bradken (Hitachi Construction Machinery)**, in a lean team where engineering, product and program roles overlapped.

CAREER PATH



SELECTED IMPACT

SmartRoller *Bradken*	Took a mining condition-monitoring product from field problem to commercial deployment: three product generations, ~5 vendors evaluated, 10+ systems built or fielded, and international field validation and deployment from Zambia to Alaska.
GET Vision *Bradken*	Recovered a delayed, vendor-led AI-vision program. Deferred a redesign that would have missed a six-month customer commitment, kept the date and used the deployment to collect field data for the next design. Deployed in Australia, Finland and Zambia.
SmartLiner *Bradken*	~10× improvement in measurement resolution on a deployed sensing product, with simpler installation.
SmartMill *Bradken*	Solved a critical RF link problem in a battery-powered rotating sensor operating inside a metal-dense mill.
Plasma-thruster electronics *Neumann Space*	Restarted power/control electronics stalled for ~6 months by diagnosing before redesigning; recovered ~1 month of program schedule while leading the electronics workstream.
ProteusONE *IBA · Japan*	Electromechanical installation and commissioning of a ~€9M proton-therapy system; the project team completed installation ~2 months ahead of a 12-month plan.

LANGUAGES

English native-level · **Hindi** native · **Japanese** basic–intermediate · **Mandarin Chinese** elementary

CAPABILITIES

Product / systems architecture	Turning field problems into requirements; feasibility; system architecture and interfaces; technical trade-offs; reusing sensing platforms across products.
Technical program leadership	Running parallel R&D programs; schedules, critical path, risks and dependencies; documented design decisions; deciding when to fix, redesign or defer; protecting customer milestones.
Electronics / RF / mechatronics	Analog, digital and mixed-signal electronics; power; RF, antennas and wireless; sensing; embedded integration; mechanical, thermal and environmental integration; controls.
Validation, productization & deployment	Bench validation and RF characterization; failure reproduction and root cause; FAT/SAT; field trials; commissioning; production readiness; handover.
Supplier & cross-functional leadership	Vendor evaluation and selection; technical scopes and acceptance criteria; supplier design reviews; aligning embedded, AI/vision, UX, mechanical, field and customer teams.

EXPERIENCE

BRADKEN (HITACHI CONSTRUCTION MACHINERY) · 2021 – Present

Mining wear-parts manufacturer building digital monitoring products for large excavators, mills and ground-engaging tools.

Principal R&D Engineer (Contract) **10/2025 – Present**
Bradken · Japan

Functional role: product systems architect, connected sensing

- Develop connected-sensing reference platforms for future mining products: custom electronics, BLE/Sub-GHz RF, low-power architecture, hardware/firmware integration and technology selection.
- Support Arctic-region deployment and U.S. market-expansion activity through Alaska field validation, commissioning and product readiness work.

Senior Mechatronics Engineer (R&D) **01/2021 – 09/2025**
Bradken · Australia

Functional role: technical, product and program lead; principal hardware/RF specialist

The R&D team was small: initially a technical manager, a mechanical engineer and this role, growing to ~6 with embedded, AI/vision, UX and junior engineers. Most detailed development was outsourced, so the technical lead also owned product definition, schedules, validation and deployment across parallel programs, and decided what stayed in-house and what went to suppliers.

- **R&D lab.** Built and ran the team's electronics, RF and mechatronics lab: PCB bring-up, RF characterization and antenna/VNA work, oscilloscope and logic analysis, power profiling, hardware/firmware integration and prototype builds. Used it to prototype independently, check supplier claims with measurements, reproduce field failures and decide when hardware was ready for FAT and the field.
- **SmartRoller — roller condition monitoring for large excavators.** Started from a field safety problem: roller temperatures were checked by hand under the undercarriage. Took it through requirements, concept and first design, evaluated ~5 vendors and technically managed the selected supplier through FAT/SAT and field trials. Over three generations, used field failures to re-architect the system and display and ran the RF/system testing, reducing installation complexity and improving serviceability on the way to commercial deployment. Led installation readiness and commissioning in Alaska; work in Australia, Zambia (RF validation), South America and Finland ranged from trials to customer deployments.
- **GET Vision — AI camera monitoring for ground-engaging tools (technical product lead, 2022–2025).** Vendor development was late, documentation was incomplete and a customer deployment was due in six months. Pieced together the architecture, open risks and dependencies, showed that a full redesign would miss the date and recommended deferring it. The deployment went ahead on time as a planned field trial, and its data shaped the redesign and productization that followed.
- **GET Vision — machine integration & global validation.** Led the work needed to make specialist vision technology run reliably on mining machines: machine integration, power, grounding, isolation, mounting, sealing, thermal, serviceability, field data and fault finding. When the vendor and internal team disagreed, brought in the internal AI specialist so the decision rested on evidence. Deployed in Australia, Finland and Zambia, where it supported a multi-million-dollar customer program.
- **SmartMill — rotating mill vibration sensing.** As technical lead and RF specialist, solved a critical link problem in a battery-powered rotating accelerometer system inside a metal-dense mill, where large data payloads had to move within short transmit windows. Coordinated the Namibia trial remotely and mentored an internal engineer to continue the electronics work.
- **GET Tracker — wireless wear monitoring.** Led product systems and hardware architecture from field problem through vendor tender, validation and Hunter Valley field trials. Proposed sharing sensing-platform elements with SmartRoller, so the work carried on after GET Tracker was discontinued.
- **SmartLiner — liner wear-sensing upgrade.** Delivered a ~10× improvement in measurement resolution on a deployed product while simplifying installation.

EARLIER EXPERIENCE

Electronics Design Engineer — 06/2020 – 12/2020
Neumann Space · Australia

Functional role: technical lead, electronics workstream — plasma-thruster power/control electronics for the SpIRIT mission

- Inherited power/control electronics that had blocked integrated testing for ~6 months. The pressure was for a quick PCB redesign; instead, stabilized the existing design first, measuring power, thermal and control behavior stage by stage until the main fault was isolated. A targeted fix put the hardware back into sustained testing and recovered ~1 month of schedule.
- Led the electronics workstream: allocated work, reviewed designs, mentored junior engineers, coordinated external specialists and vendors, and escalated technical risk to management with measurement evidence. Contributed to advancing the propulsion electronics from TRL 4 toward TRL 6.

Robotics / Electronics Engineer (Contract) — 03/2020 – 01/2021
AML3D · Australia — full-time, then part-time alongside Neumann Space

- Re-architected the electrical and control system of the ARCEMY robotic wire-arc additive manufacturing (WAAM) platform across PLC/SCADA and ABB RAPID, improving fault handling, repeatability and maintainability.
- Aligned controls with manufacturing workflow, operating procedures and quality-system requirements, helping move a research-grade system toward repeatable customer-facing operation.

RF Systems Engineer — 07/2019 – 12/2019
Mine Site Technologies (MST) · Australia

- Designed underground and surface RF communication and tracking systems: architecture input, BOM, installation planning for site-specific propagation and regulatory constraints, commissioning and acceptance testing.

Electromechanical Engineer — 02/2017 – 06/2019
Ion Beam Applications (IBA) · Toyohashi, Japan — ProteusONE proton-therapy system (~€9M)

- Joined at the early site phase and worked through rigging, installation, alignment, commissioning, acceptance and handover of a superconducting synchrocyclotron (230 MeV), rotating gantry and 6-DOF patient-positioning robot, with high-power electrical, RF, vacuum, cryogenic, cooling, PLC/SCADA and imaging interfaces.
- Responsible for electromechanical installation quality: supervised contractor work within scope, rejected substandard work and required correction, then tested, documented and released subsystems into later testing.
- Diagnosed interacting mechanical, brake, control and alignment faults on the patient-positioning robot and fed commissioning evidence back to design engineering. Part of the team that completed installation in ~10 months against a ~12-month plan; stayed through the center's first patient treatment (09/2018) and trained the local team for handover.

Electronics / RF Engineer — 09/2015 – 01/2017
Trans Communication · Australia

- Performed RF fault isolation, calibration, ACMA compliance testing and commissioning across radio front-ends, IF chains, amplifiers and antenna systems.

Earlier: Service Engineer, Pancham Enterprises (India, 2010–2011) — control valves and compressor turbines; Customer Service Engineer, Siemens (India, 2010) — building-systems commissioning and maintenance.

EDUCATION

Master of Engineering Management *University of Queensland*	Electronics & Electrical Engineering · 2011 – 2013
Bachelor of Technology *Jawaharlal Nehru Technological University*	Electronics & Communication Engineering · 2006 – 2010
Professional Development *University of Queensland*	Antenna Design · 2015

TECHNICAL TOOLKIT

Test & measurement	VNA, spectrum analyzer, oscilloscope, logic analyzer, power profiling, electronic loads, rework and inspection.
Wireless, interfaces & controls	BLE, Sub-GHz/ISM, LTE, GNSS, MQTT; CAN, RS485/RS232; STM32, TI CC13xx; PLC/SCADA, ABB RAPID, industrial robotics.
Design & analysis	Altium, KiCad, LTspice, ANSYS HFSS / Mechanical; Python, MATLAB, C/C++; Jira, SAP.